

StudyNest

Project Specification & Product Blueprint

Tagline: Organize your learning. Track your progress. Build better study habits.

Portfolio positioning: A polished full-stack student productivity platform demonstrating product thinking, CRUD, authentication, relational database design, analytics, responsive UI/UX, accessibility, testing, and deployment.

1. Project Overview

StudyNest is a modern study management platform that gives students one focused workspace for courses, assignments, study sessions, learning goals, bilingual notes, and progress analytics.

Its features form a coherent workflow: create a course, define goals, add tasks, study, record sessions, write notes, and use analytics to understand progress.

Core workflow: Create Course → Create Goals → Add Tasks → Study → Record Session → Write Notes → Track Progress → View Analytics.

2. Problem & Target Users

Students often spread learning information across task apps, notebooks, calendars, timers, and spreadsheets. StudyNest brings the essential workflow into one focused product.

Target users: university students, self-learners, programming learners, language learners, and exam-preparation students.

3. Main Features

Authentication

- Register, login, logout, password reset, profile management, protected routes, session persistence.

Dashboard

- Study overview, study-time summary, task summary, active goals, courses, weekly activity, and today's tasks.

Course Management

- Complete CRUD, descriptions, colors, target dates, status, and dedicated course pages.

Tasks & Assignments

- CRUD, status, priority, due dates, estimated time, search, filtering, sorting, and course association.

Study Session Tracker

- Start/pause/resume/finish timer, course association, duration storage, focus rating, session notes, and time summaries.

Bilingual Notes

- English and Chinese content, course association, tags, search, filtering, editing, and deletion.

Learning Goals

- Goal creation, target/current values, deadlines, progress visualization, status, and course association.

Calendar

- Simple monthly calendar showing assignments, study sessions, and goal deadlines.

Analytics

- Study time charts, task completion, course distribution, goal progress, weekly activity, average session length, and completion rate.

Global Search

- Search across courses, tasks, and notes with grouped results.

UX & Accessibility

- Loading/empty/error states, toasts, confirmation dialogs, keyboard access, semantic HTML, focus states, labels, and responsive behavior.

Themes

- Professional light theme and complete dark mode with persisted preference.

4. Pages & Navigation

Public: Landing, Login, Register. Authenticated: Dashboard, Courses, Course Details, Tasks, Notes, Goals, Calendar, Analytics, Settings.

Desktop uses a persistent sidebar; mobile uses bottom navigation or a compact responsive menu. The interface should feel like focused productivity SaaS software, not a generic template.

5. UI/UX Direction

Visual direction: professional productivity software inspired by the clarity of Linear and Notion without copying either visual identity.

Principles: hierarchy over decoration, restrained color, subtle borders/shadows, consistent spacing, clear states, accessibility, and responsive-first design.

Avoid excessive glassmorphism, giant rounded cards, heavy gradients, and generic AI-dashboard styling.

6. Color System

Token	Color	Usage
Primary	#172033	Navigation, headings, important text
Background	#F8F9F7	Main light-mode background
Surface	#FFFFFF	Cards, panels, forms

Token	Color	Usage
Accent	#16A36A	Primary actions, progress, active states
Soft Accent	#DDF5E9	Selected/soft highlights
Secondary Text	#687386	Supporting copy
Muted	#9AA3B2	Metadata and disabled text
Danger	#E45D5D	Destructive/error states

7. Dark Theme

Token	Color	Usage
Background	#10141C	Main background
Surface	#171D27	Cards and panels
Border	#28303D	Dividers and inputs
Text	#F3F5F7	Primary text
Muted	#929BAA	Secondary text
Accent	#35C98A	Actions and progress

8. Typography & Components

Font: Inter. Headings 600–700; body 400; labels 500.

Components: buttons, inputs, badges, tabs, dialogs, dropdowns, charts, skeletons, empty states, and toasts.

Cards: around 12px radius, subtle 1px border, restrained shadows. Icons: Lucide.

9. Database Design

Primary tables: profiles, courses, tasks, notes, goals, study_sessions.

Table	Important fields
profiles	id, user_id, name, avatar_url, bio, created_at, updated_at
courses	id, user_id, title, description, color, target_date, status, created_at, updated_at
tasks	id, user_id, course_id, title, description, status, priority, due_date, estimated_minutes, created_at, updated_at
notes	id, user_id, course_id, title, english_content, chinese_content, tags, created_at, updated_at
goals	id, user_id, course_id, title, description, target_value, current_value, deadline, status, created_at, updated_at
study_sessions	id, user_id, course_id, started_at, ended_at, duration_minutes, focus_rating, description, created_at

Relationship model: User → Courses → Tasks/Notes/Goals/Study Sessions. Every user's private data must be isolated with database-level security policies.

10. Technology Stack

Layer	Technology
Frontend	Vue 3 + TypeScript + Vite
Styling	Tailwind CSS
State	Pinia
Routing	Vue Router
Backend / Platform	Supabase
Database	PostgreSQL
Authentication	Supabase Auth
Security	PostgreSQL Row Level Security
Charts	Chart.js
Icons	Lucide Icons
Validation	Zod
Utilities	VueUse, date-fns
Deployment	Vercel + Supabase
Version Control	Git + GitHub

11. Recommended Architecture

```

src/
├── assets/
├── components/
│   ├── ui/
│   ├── dashboard/
│   ├── courses/
│   ├── tasks/
│   ├── notes/
│   └── analytics/
├── layouts/
├── pages/
├── stores/
├── services/
├── types/
├── composables/
├── utils/
└── router/

```

12. Development Roadmap

Phase	Focus	Deliverable
1	Foundation	Vue, TypeScript, Vite, Tailwind, Router, Pinia, Supabase, ESLint, Prettier.
2	Authentication	Register/login/logout, protected routes, persistence, profile.
3	Database	Schema, relationships, constraints, indexes, RLS policies.
4	Courses	Complete course CRUD and detail page.

Phase	Focus	Deliverable
5	Tasks	CRUD, status, priority, deadlines, search, filtering, sorting.
6	Notes	Bilingual content, tags, search, filtering, CRUD.
7	Study Sessions	Timer, pause/resume, storage, duration calculations.
8	Goals	Targets, progress, deadlines, status.
9	Dashboard	Real-data summaries, tasks, goals, sessions, course progress.
10	Analytics	Charts and derived metrics.
11	UX Polish	States, toasts, dialogs, accessibility, responsiveness, dark mode.
12	Testing & Deployment	Critical tests, production build, deployment, auth/database verification.

13. MVP vs Version 2

MVP: Authentication, Courses, Tasks, Notes, Goals, Study Sessions, Dashboard, Analytics, Dark Mode, Responsive UI.

Version 2: Pomodoro mode, streaks, recurring tasks, calendar integration, Markdown notes, attachments, notifications, PWA/offline support, data export.

14. Portfolio & CV Positioning

Portfolio: StudyNest is a full-stack study management platform for organizing courses, assignments, study sessions, bilingual notes, learning goals, and progress analytics.

CV: Emphasize Vue 3 + TypeScript, Supabase Auth/PostgreSQL RLS, reusable Vue/Pinia architecture, CRUD/search/filter workflows, analytics dashboards, and responsive SaaS UI.

15. Interview Topics

- Vue reactivity, component architecture, routing, state management, forms, validation, and error/loading states.
- PostgreSQL relationships, constraints, indexes, RLS, and user data isolation.
- Authentication flow, session persistence, and protected routes.
- Architecture decisions and separation of components, stores, services, and types.
- Product decisions: why bilingual notes, why study-session tracking, and what could be built next.

16. Definition of Success

- A new user can register and reach a useful dashboard without confusion.
- Users can create courses and manage related tasks, notes, goals, and study sessions.
- Progress metrics are calculated correctly from real database data.
- Users cannot access another user's private data.
- Desktop and mobile experiences are polished and consistent.

- The project is deployed and documented with screenshots, architecture, schema, setup instructions, and lessons learned.